Naresh S

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PROFESSIONAL SUMMARY

Pre-final year Integrated M.Tech Software Engineering student at VIT Chennai with strong interest in Machine Learning, Deep Learning, and AI/ML model development. Experienced in building CV, NLP, and Generative AI systems using PyTorch, HuggingFace Transformers, and YOLOv8. Skilled in problem-solving, model evaluation, and developing scalable AI solutions.

EDUCATION

Vellore Institute of Technology, Chennai, India 2022 - 2027
Integrated M.Tech Software Engineering - CGPA: 8.28
Jayam Vidhyalaya Matric Higher Secondary School 2020 - 2022
SSLC - 97.4% HSC - 91.5%

TECHNICAL SKILLS

Programming: Python, Java, C++, SQL
ML/DL Frameworks: PyTorch, TensorFlow, Scikit-Learn, HuggingFace Transformers, YOLOv8 (Ultralytics)
AI/ML: Computer Vision, NLP (NER, NLU, NLG), Generative AI, Transfer Learning, Object Detection
Backend & Tools: Flask, REST APIs, Docker (basic), Git, GitHub, Jupyter Notebook, Power BI
IoT & Embedded: Arduino, ESP32, Raspberry Pi, MQTT, CoAP
Core Concepts: OOP, DBMS, OS, DSA, Statistical Analysis, Software Architecture

PROJECTS

Document Layout Classifier for Media Publishers | PyTorch, ResNet18, Transfer Learning, PubLayNet

- Built a CNN classifier using PyTorch and ResNet18 transfer learning to classify document page regions into 5 layout types (article, table, figure, advertisement, mixed), directly applicable to automated content workflows for media publishers.
- Implemented custom classifier head with Dropout regularisation, StepLR scheduling, and data augmentation (RandomCrop, ColorJitter) to prevent overfitting on limited training data.
- Evaluated using macro F1 score and confusion matrix; saved best checkpoint via early stopping for production-ready inference.

Pharma Packaging NER & Compliance Checker | HuggingFace Transformers, BioBERT, BIO Tagging, seqeval

- Developed a biomedical NER system using BioBERT to extract 6 pharma-specific entities (drug, dosage, route, frequency, warning, manufacturer) from packaging label text, with BIO tagging and WordPiece subword-label alignment.
- Built an automated compliance checker that verifies presence of all regulatory-required entities and outputs PASS/FAIL verdicts per label, mirroring pharma packaging artwork compliance workflows.
- Evaluated using entity-level F1 score via seqeval; demonstrated BioBERT's advantage over plain BERT on biomedical domain terminology.

AI Content Quality Checker | Generative AI, LLM, Structured Prompting, HuggingFace

- Designed an LLM-powered editorial auditor using zero-shot structured JSON prompting to assess news articles for clarity score, tone (neutral/biased/sensationalist), 5W completeness, and flagged sentences with suggested rewrites.
- Engineered system/user prompt separation with strict JSON schema output constraints and greedy decoding (temperature=0) to ensure consistent, parseable results and minimise hallucination.
- Implemented modular backend supporting HuggingFace Transformers and Ollama for fully local, API-independent deployment.

Pharma Packaging Defect Detector | YOLOv8, Ultralytics, OpenCV, Synthetic Data Generation

- Fine-tuned YOLOv8n to detect 4 packaging defect classes (text blur, label misalignment, missing elements, print artifacts) with per-defect severity classification (HIGH/MEDIUM/LOW) and PASS/WARN/FAIL verdict output.
- Built a synthetic training data pipeline using Pillow generating 300+ annotated packaging images with programmatic defect injection, bypassing the lack of public pharma packaging datasets.
- Implemented full inference pipeline with OpenCV bounding box visualisation; evaluated using mAP@0.5 with YOLOv8 built-in augmentation (mosaic, mixup) and early stopping.

IoT Monitoring Dashboard with Testing | Python, MQTT, ThingSpeak, Pytest

- Developed an IoT system to collect and monitor real-time sensor data via MQTT with ThingSpeak cloud storage and dashboard visualisation.
- Designed Pytest test cases to validate data flow, system responses, and edge cases; performed functional testing and debugging to ensure system reliability.

PUBLICATIONS

Siddha Medical Dataset for Research Applications IEEE DataPort
DOI: 10.21227/qkyv-5j98

- Published a structured healthcare dataset to support AI/ML research.
- Collected, cleaned, and organized Siddha medical data for academic use.
- Designed dataset architecture to support healthcare analytics and machine learning studies.

CERTIFICATIONS

Oracle AI Associate | Agile Scrum in Practice